Exploring Machine Learning Optimizatios for Pair Selection in Pairs Trading

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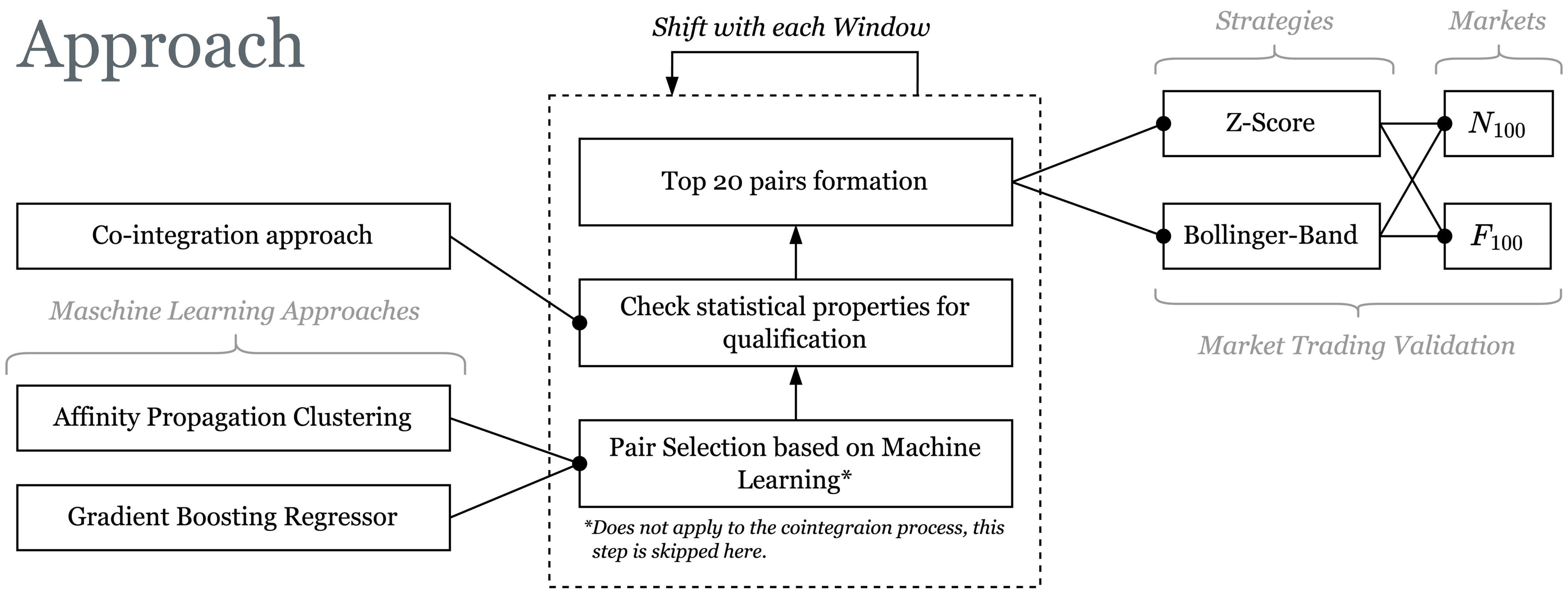
Approach



Three pair identification methods (traditional cointegration, Affinity Propagation Clustering, and Gradient Boost Regressor) are compared using Engle-Granger validation and top 20 pair selection. A 12-month sliding window approach tests all methods with Z-Score and Bollinger Bands strategies on NASDAQ-100 and FTSE-100 markets.

Affinity Propagation Clustering:

The method automatically determines optimal cluster numbers by grouping stocks with similar **return** and **volatility** profiles. The economic assumption is that stocks with comparable risk-return characteristics respond similarly to market conditions, creating more stable statistical relationships. Clustering reduces the search space from thousands to a few hundred pair combinations within clusters, with final pair selection based on **proximity to cluster centers** and **profile similarity**.

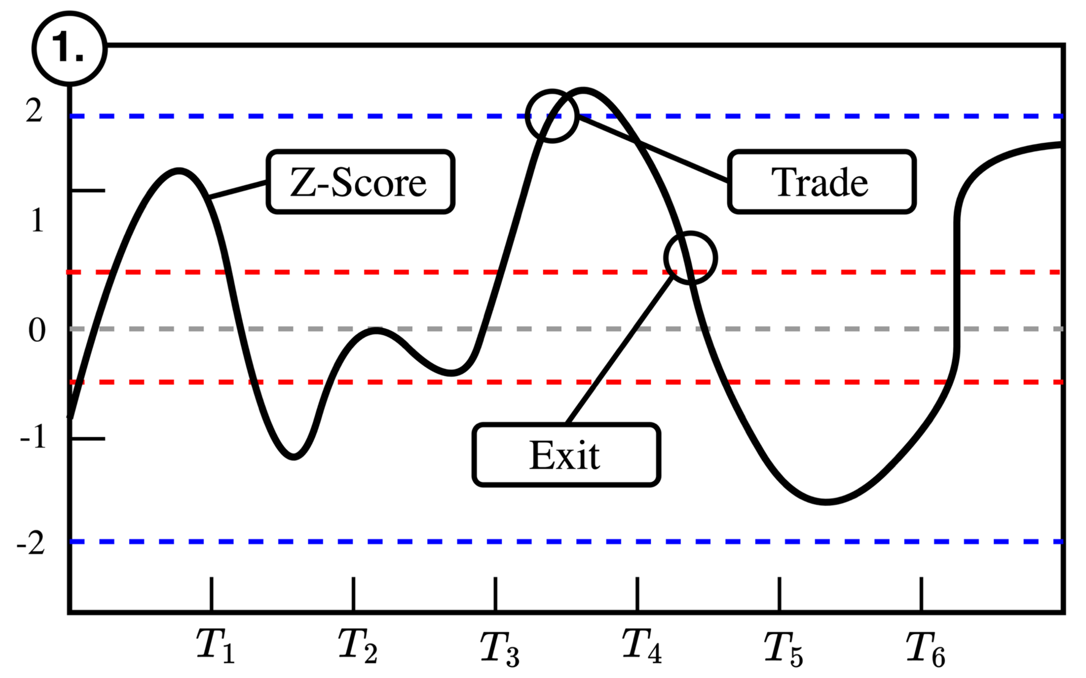
Gradient Boost Regressor:

The model combines multiple weak decision trees to predict average **spread deviations between stock pairs**, essentially forecasting which pairs will have minimal future spread volatility. Instead of trading all cointegrated pairs and waiting for convergence, the model selects only pairs with predicted spread stability. It uses a comprehensive feature set including **correlation measures**, **technical indicators** like Williams Oscillators and Bollinger Bands, volume characteristics, and momentum indicators to make these predictions.

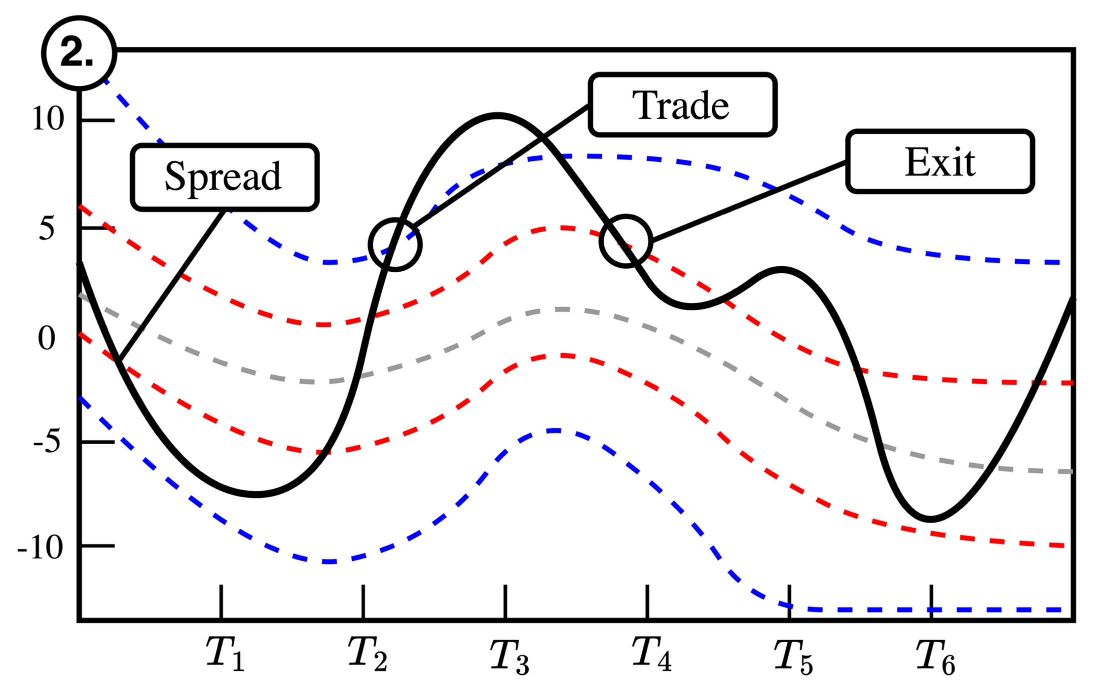
Details & Backtester

tinyurl.com/pairtrade

Strategies

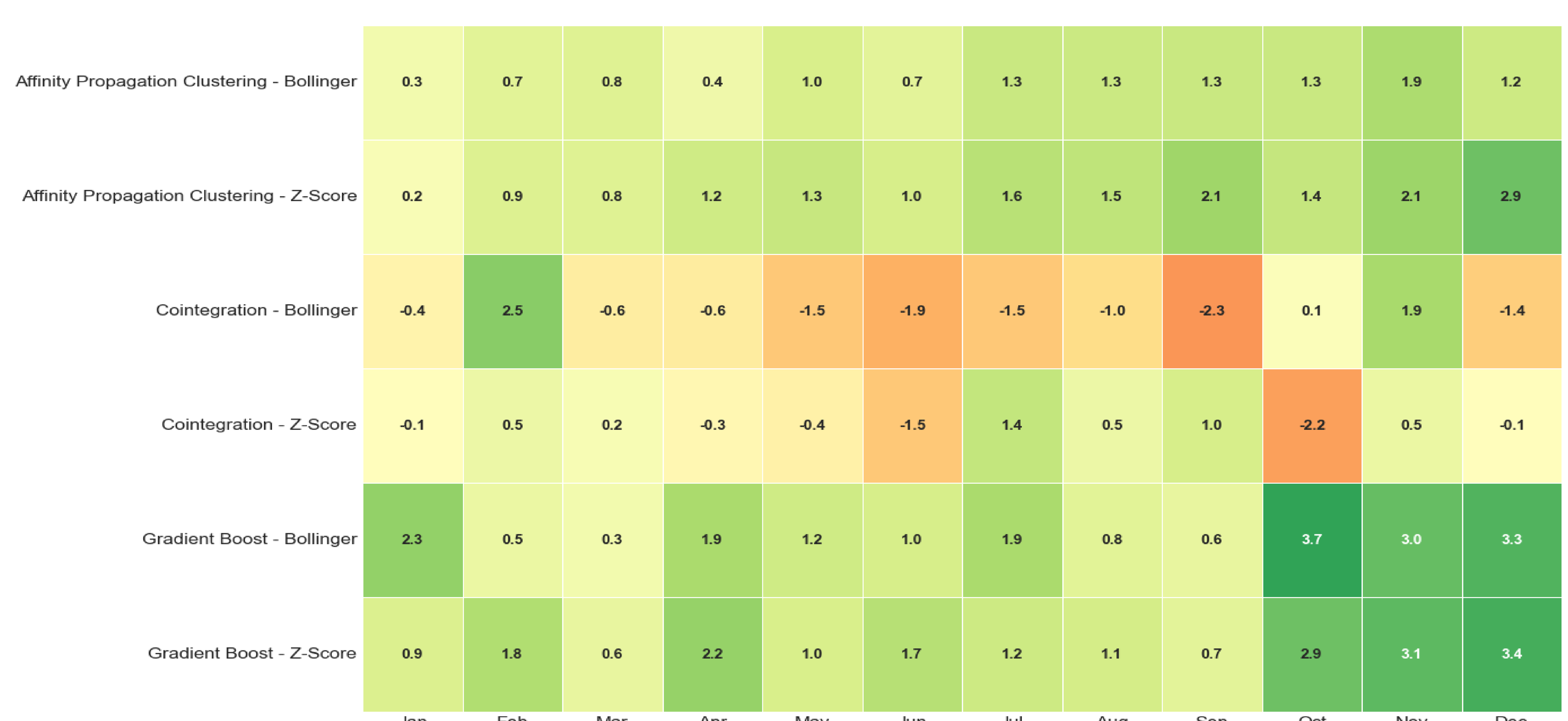


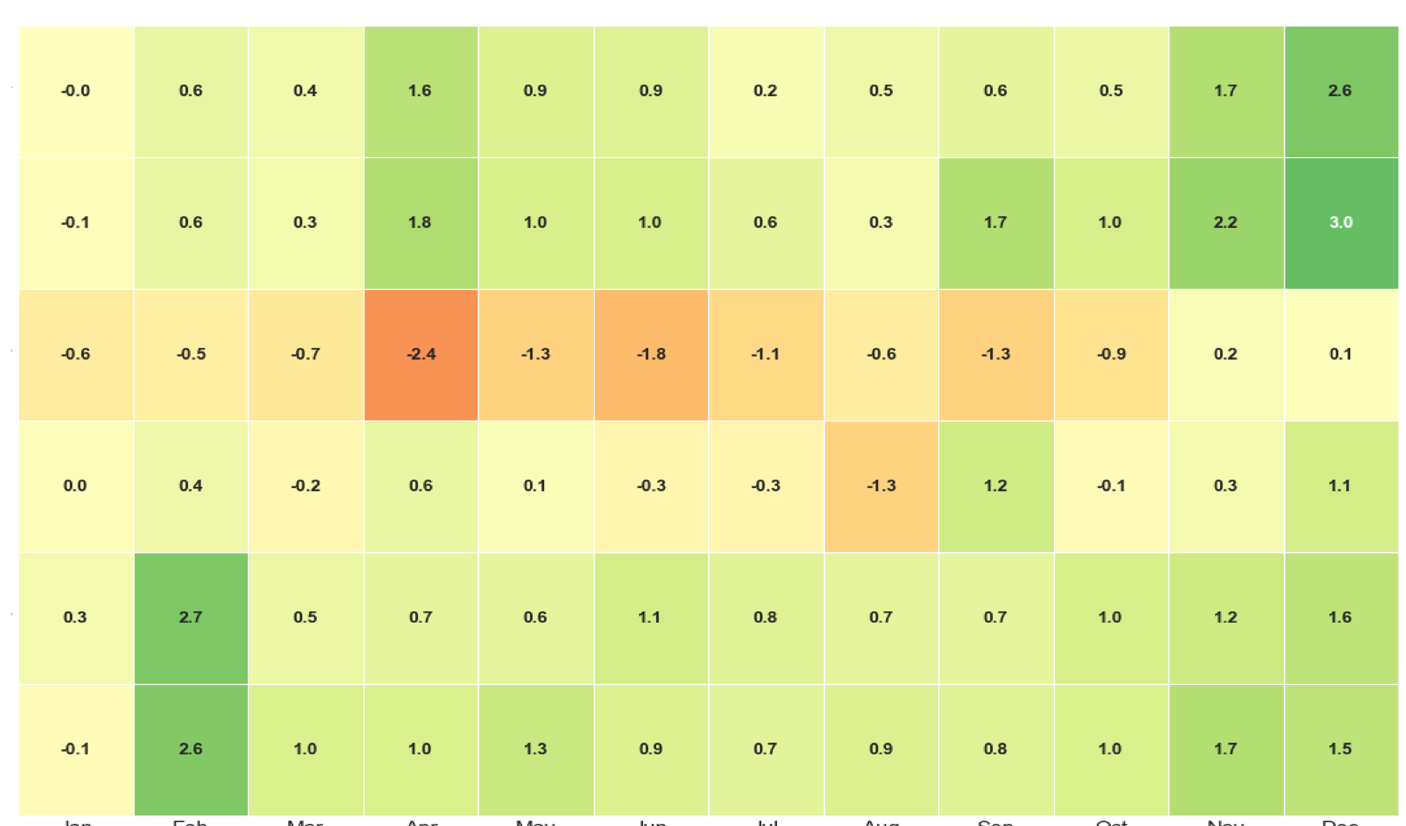
Z-Score: Enters at fixed ±2.0 thresholds, exits when spread returns to zero.



Bollinger Bands: Uses adaptive volatility bands with dynamic hedge ratios for entry/exit signals

Results (NASDAQ100 left, FTSE100 right)





Ansatz	Strategy	Return (%)	Max Drawdown (%)
Affinity Propagation	Z-Score	14.30	-0.09
Gradient Boost	Z-Score	13.75	-0.09
Gradient Boost	Bollinger	12.11	-0.09
Affinity Propagation	Bollinger	11.18	-0.07
Kointegration	Z-Score	0.81	-0.88
Kointegration	Bollinger	-12.57	-13.09

Ansatz	Strategy	Return (%)	Max Drawdown (%)
Gradient Boost	Z-Score	20.90	-0.11
Gradient Boost	Bollinger	20.47	-0.18
Affinity Propagation	Z-Score	17.98	-0.09
Affinity Propagation	Bollinger	13.53	-0.05
Kointegration	Z-Score	-1.13	-3.87
Kointegration	Bollinger	-11.21	-11.99